SOFTWARE ENGINEERING

Forensic Data Recovery

for XFS & Btrfs File Systems

An automated, verifiable recovery toolkit that helps investigators pull deleted files and metadata back from modern Linux file systems.



Background

Modern Linux systems, NAS appliances and cloud storage backends have largely moved on from FAT and NTFS. XFS is the default file system on several major enterprise Linux distributions because of its scalability, and Btrfs is used wherever copy-on-write snapshots, checksummed data and flexible volume management matter — think Synology and QNAP NAS boxes, container hosts and modern desktop distributions.

That shift matters for digital forensics. When one of these devices is deleted, corrupted, or seized as evidence, investigators still reach for recovery tools designed decades ago for far simpler file systems — and those tools were never built to understand journaling, B-trees, or copy-on-write metadata.



Where XFS & Btrfs show up

- Enterprise Linux servers (RHEL & derivatives)
- NAS appliances (Synology, QNAP)
- Container & cloud storage hosts
- Modern desktop Linux distributions

Problem Statement

There is currently no easy-to-use, automated, and verifiable recovery system for these file systems.

Fragmented tooling

Existing utilities cover only one OS or one file system at a time, forcing investigators to juggle several tools per case.

Slow & manual

Recovery today leans on manual, command-line-heavy workflows that assume deep file-system expertise.

No verified integrity

Few tools prove — with checksums or hashes — that recovered data is complete and untampered.

Not evidence-ready

Outputs rarely come with the structured, exportable reports that legal and case-documentation workflows need.

Project Objectives

Five goals guide every design decision in this toolkit.

Recover with accuracy

Improve the accuracy of forensic data recovery from XFS and Btrfs file systems, down to fragmented and partially overwritten files.

Automate the hard part

Provide automated recovery of deleted files and their associated metadata, without hand-written scripts per case.

Stay usable

Offer an intuitive, cross-platform tool that simplifies file recovery for both investigators and general users.

Prove integrity

Ensure recovered data can be trusted through checksum validation and hash-based verification against source blocks.

Support the case file

Support forensic investigations with comprehensive, exportable reports for documentation and case analysis.

Proposed Approach

A four-layer architecture — each layer owned by tools built for that exact job.



Presentation Layer

Cross-platform desktop GUI for selecting a drive, watching the scan progress live, previewing recovered files, and downloading the final report.



Recovery Engine

Parses XFS B-trees and Btrfs copy-on-write metadata to walk free blocks, rebuild file records, and extract deleted files with their metadata intact.



Verification Layer

Hashes every recovered file and compares it against source block checksums, so nothing is reported as recovered unless it is provably intact.



Reporting & Storage Layer

Logs case metadata locally and compiles a structured, exportable report ready for documentation or legal review.

Technology Stack

Built entirely on proven, freely available forensic and cross-platform tooling.



Python 3

Core recovery logic, orchestration & CLI/API layer



C / C++

Low-level disk I/O & performance-critical parsing



The Sleuth Kit (TSK)

Forensic primitives & disk-image handling



PyQt6 / Electron.js

Cross-platform desktop GUI (Win / macOS / Linux)



SQLite

Lightweight local case & metadata storage



hashlib (SHA-256)

Checksum-based integrity verification



ReportLab / WeasyPrint

Automated PDF / HTML forensic reports



Git & GitHub Actions

Version control and continuous integration

Key Features

Seven capabilities that make this toolkit practical for real casework.



Cross-platform compatibility



Comprehensive file recovery



Advanced metadata extraction



Checksum & hash verification



User-friendly GUI



Automated reporting



Efficient scanning algorithm

Innovation & Uniqueness

What sets this toolkit apart from typical recovery utilities.

Cross-platform tool

Works across Linux, Windows, macOS and embedded systems from a single codebase.

Specialized focus

Purpose-built for advanced recovery on XFS and Btrfs, where most tools fall short.

Intuitive GUI

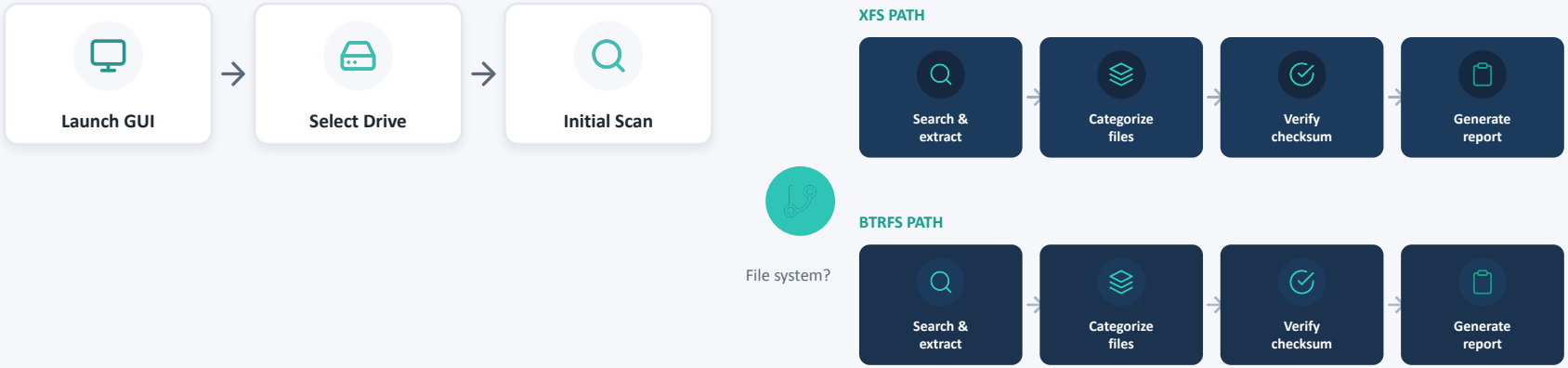
A friendly interface for navigating, previewing and recovering files without CLI expertise.

Scalability

Handles everything from a single USB drive to large-scale enterprise storage.

System Workflow

One guided workflow, with the recovery engine branching by file system underneath.



Both paths converge on a single, user-facing result screen with a recovery summary and downloadable report.

End of workflow

Impact & Benefits

Practical gains for investigators today, and a foundation for tomorrow.

Benefits

- Time efficiency in recovery workflows
- Cost-effective alternative to commercial suites
- Comprehensive, user-friendly reporting
- Real-time analysis as scans progress
- Documentation ready for legal verification
- Scalable to future storage needs
- Encourages community engagement & contribution

Impacts

- Improved forensic outcomes in casework
- Strengthened legal procedures & evidence trails
- Increased efficiency in investigations
- Adaptation to fast-changing file system technology
- Meaningful contribution to cybersecurity efforts
- Strengthened legal processes across agencies



Thank You

Forensic Data Recovery for XFS & Btrfs File Systems

Questions & feedback welcome.